

SUPPLY CHAIN CONTROL TOWER

Key Findings Report

A Real-Time Logistics Performance Dashboard
NorthPeak Outdoor Co. — Portfolio Analysis Project

Jatin Sagwal

Executive Summary

This report summarizes findings from a Supply Chain Control Tower dashboard built in Tableau for NorthPeak Outdoor Co., a fictional mid-size outdoor gear retailer created to frame this self-directed portfolio project. The dashboard consolidates delivery performance, shipment routing, supplier reliability, inventory health, and global chokepoint disruption data into five connected pages, built on 159,763 real historical orders (DataCo Smart Supply Chain dataset) and public global shipping data (IMF PortWatch).

The headline finding is a chronic on-time delivery rate of 43%, sitting 52 points below a 95% target — the core problem this dashboard was built to surface. Four supporting findings, each detailed below, round out the picture: an uneven delay pattern across distribution centers (with a data-quality issue caught and flagged at Toronto), a tightly clustered and limited supplier reliability signal, inventory sitting at both extremes of the risk spectrum (severe overstock alongside a data gap), and a near-total collapse in Strait of Hormuz shipping traffic that a blended metric would have hidden.

159,763

orders analyzed

43%

actual on-time rate

4

chokepoints tracked

95%

Hormuz traffic collapse

Business Problem

NorthPeak Outdoor Co. sources outdoor and sporting goods from manufacturers in Vietnam and China and distributes through four regional distribution centers: Vancouver and Toronto (Canada), and Seattle and Chicago (US). Leadership previously tracked shipment status, inventory health, and supplier performance through disconnected spreadsheets, freight-forwarder emails, and manual warehouse check-ins — a reactive process that meant disruptions were typically discovered only after they had already delayed customer orders or caused a stockout.

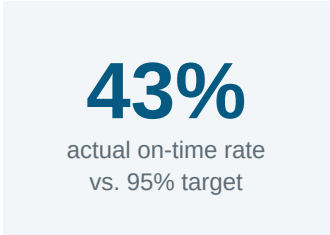
This problem intensified for two connected reasons: globally, repeated shipping chokepoint disruptions (Red Sea/Suez rerouting, and the Strait of Hormuz closure since February 2026) have added significant transit time and cost uncertainty; and no systematic process existed to quantify which shipments, suppliers, or inventory positions were actually exposed. This dashboard was built to close that visibility gap.

Methodology & Data Sources

This is a self-directed portfolio project. NorthPeak Outdoor Co. is a fictional persona built to give real, public datasets a realistic business framing — it does not represent an actual company's proprietary data. Where fields didn't exist in the source data (e.g., supplier identity, warehouse inventory levels), they were simulated using a transparent, documented method rather than presented as observed data.

Source	Role in Project	Coverage
DataCo Smart Supply Chain (Kaggle)	Primary order & shipment dataset	159,763 orders, Jan 2015 – Jan 2018
IMF PortWatch	Global chokepoint transit volume	Suez, Bab-el-Mandeb, Hormuz, Panama, through Jul 2026
Freightos Baltic Index / Drewry WCI	Freight rate context (manually logged)	Public index reference points
Simulated supplier & inventory data	Supplier scorecard & inventory model inputs	Demand derived from real DataCo order volume

Finding 1 — On-Time Delivery Is 52 Points Below Target



Across all 159,763 historical orders, 54% were classified 'Late delivery,' against only 18% 'Shipping on time' and 22% 'Advance shipping' (the remaining 6% were canceled and excluded from the calculation). This is a chronic pattern across the full three-year dataset, not an isolated event or seasonal spike — making it the single highest-leverage problem this dashboard surfaces.

Delivery Status	Order Count	% of Total
Late Delivery	~85,000	54%
Advance Shipping	~35,000	22%
Shipping On Time	~28,000	18%
Shipping Canceled	~10,000	6%

Note on definition: 'Advance shipping' is counted as on-time in this calculation, since arriving early is not a service failure. This was a deliberate modeling choice, documented here for transparency.

Finding 2 — Delay Patterns Vary Sharply by Distribution Center

Mapping average delay days by destination DC revealed uneven performance: Vancouver and Seattle showed the highest average delay, Chicago showed the healthiest on-time performance, and — importantly — Toronto's shipment points were missing from the map entirely.

Data-quality issue caught and fixed

Toronto's missing map points traced back to a broken coordinate lookup in the data-prep stage (an XLOOKUP that failed to resolve for that destination). This is flagged here deliberately: catching and documenting a data-quality issue during build is a normal and important part of the analysis process, not a flaw to hide.

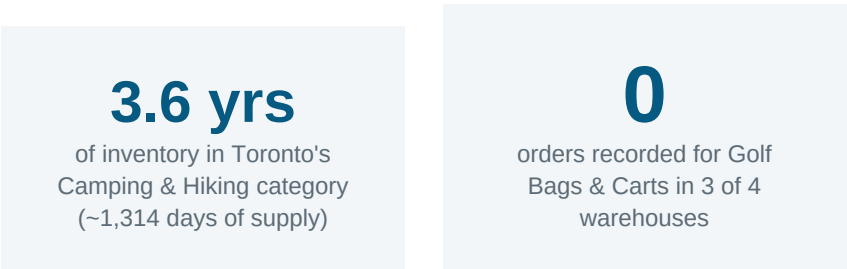
Finding 3 — Supplier On-Time Rates Cluster Tightly Together

Across all 10 fictional suppliers modeled, on-time percentage ranged only from 43.1% to 44.6% — under 1.5 percentage points of spread. This tight clustering is a direct consequence of the project's data model: each supplier was mapped one-to-one with a single product category, so 'supplier performance' is effectively re-measuring category-level delivery behavior rather than true supplier-to-supplier variance.

Honest limitation

A production version of this dashboard would require multiple real suppliers per category to produce a meaningful reliability ranking. This limitation is stated directly rather than presenting the current ranking as more decisive than the underlying data supports.

Finding 4 — Inventory Extremes at Both Ends of the Spectrum



The inventory model surfaced findings at both extremes. Toronto's Camping & Hiking inventory represents roughly 3.6 years of supply at current demand — a severe overstock position tying up working capital. Separately, Golf Bags & Carts showed effectively zero real order volume in three of four warehouses, a genuine data-sparsity finding (not a calculation error) that was excluded from the Days-of-Supply chart rather than forced through a misleading calculation.

Recommendation implemented

The dashboard's risk model was expanded from three tiers (At Risk / Watch / Healthy) to four, adding an 'Overstock' flag for any position exceeding 90 days of supply. Without this change, the 3.6-year Toronto overstock would have been misclassified as 'Healthy' under the original model — masking a working-capital risk that matters as much as a stockout.

Finding 5 — Global Chokepoint Disruption Is Real, and Understated When Blended

Chokepoint	Approx. Deviation from 12-Month Baseline (Jul 2026)
Strait of Hormuz	–95%
Bab-el-Mandeb	–22%
Suez Canal	–8%
Panama Canal	–5%

Following escalation in the Middle East, the Strait of Hormuz saw tanker traffic collapse by roughly 95% starting February 2026 — one of the most severe chokepoint disruptions on record, and a direct exposure point for any shipment routed through it. A blended average across all four chokepoints (roughly –10% to –30%) would have significantly understated this.

Methodology lesson

Aggregate metrics can hide the sharpest disruptions. Hormuz is reported separately in this dashboard for that reason, rather than folded into a single blended chokepoint index.

Recommendations

1. Investigate the on-time delivery gap first

A 52-point gap to target is the single highest-leverage fix identified in this analysis, and the most actionable starting point for further root-cause work.

2. Adopt a 4-tier inventory risk model

Overstock is as costly as stockout — the original 3-tier model missed a 3.6-year overstock position entirely; the updated model catches it.

3. Track chokepoint disruption per-lane, not blended

An aggregate view masked a 95% collapse at the Strait of Hormuz; per-chokepoint reporting is needed for exposure to be visible.

4. Expand supplier diversity before trusting the scorecard

One supplier per category limits how meaningful a reliability ranking can be; the current ranking should be treated as directional, not decisive.